

Qualification Pack



Welding

QP Code: ASC/Q3111

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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ASC/Q3111: Welding

Brief Job Description

A welder prepares and joins a range of metals and metallic alloys using mainly processes where an electric arc is the heat source. Electric arc processes utilize a gas shield or a flux to protect the molten weld area from contamination by the surrounding atmosphere. A welder needs to be able to interpret engineering drawings, standards and symbols and correctly translate these requirements into accurate structures and fabrications.

Personal Attributes

The person should be patient, organised, team-oriented and have the ability to work for long hours in adverse conditions. They should be keen observers and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N9819: Work organization and management \(Welding\)](#)
2. [ASC/N3120: Preparation and assembly techniques](#)
3. [ASC/N3121: Welding materials](#)
4. [ASC/N3122: SMAW \(111\) and GMAW \(135\) Process](#)
5. [ASC/N3123: FCAW-G \(136\) Process](#)
6. [ASC/N3124: GTAW \(141\) Process](#)
7. [ASC/N3125: Finishing, quality assurance, and testing](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
Country	India
NSQF Level	4.5

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Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7212.0302
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	22 Years
Last Reviewed On	NA
Next Review Date	30/06/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-AU-01811-2024-V1-ASDC
NQR Version	1

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ASC/N9819: Work organization and management (Welding)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Manage work and resources

Elements and Performance Criteria

Manage work and resources

To be competent, the user/individual on the job must be able to:

- PC1.** Work safely with regard to themselves and others
- PC2.** Select, wear, and maintain PPE as required
- PC3.** Recognize hazardous situations and take appropriate actions with regard to their own and others' safety
- PC4.** Follow correct procedural processes when working in hazardous environments
- PC5.** Locate and identify dimensions and weld symbols
- PC6.** Adhere to manufacturers' safety data sheets
- PC7.** Maintain a clean working environment
- PC8.** Complete work within agreed timescales
- PC9.** Make essential connections for specific welding procedures
- PC10.** Problem solve efficiently
- PC11.** Take account of the physical and mental demand of the tasks
- PC12.** Use hand and eye coordination with precision
- PC13.** Utilize fume extraction systems efficiently to remove emissions
- PC14.** Create efficient work practices to eliminate waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Standards and legislation relating to the health, safety, security, and hygiene in the welding industry
- KU2.** Range, use and maintenance of personal protective equipment used in the industry for any given circumstances
- KU3.** Selection and use of safety equipment related to specific or hazardous tasks
- KU4.** ISO A and/or E (American and European) drawing representation
- KU5.** Technical terms and symbols used in drawings and plans

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- KU6.** Terminology and safety data supplied by manufacturers
- KU7.** Requirements and effects of welding production for the environment and sustainability issues
- KU8.** Basic mathematical manipulation and unit conversion
- KU9.** Geometrical principles, techniques, and calculations
- KU10.** Time management

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage work and resources</i>	30	50	-	20
PC1. Work safely with regard to themselves and others	2	4	-	2
PC2. Select, wear, and maintain PPE as required	2	3	-	1
PC3. Recognize hazardous situations and take appropriate actions with regard to their own and others' safety	3	5	-	2
PC4. Follow correct procedural processes when working in hazardous environments	3	5	-	2
PC5. Locate and identify dimensions and weld symbols	2	3	-	2
PC6. Adhere to manufacturers' safety data sheets	2	3	-	2
PC7. Maintain a clean working environment	2	4	-	1
PC8. Complete work within agreed timescales	2	3	-	1
PC9. Make essential connections for specific welding procedures	2	3	-	2
PC10. Problem solve efficiently	2	3	-	1
PC11. Take account of the physical and mental demand of the tasks	2	3	-	1
PC12. Use hand and eye coordination with precision	2	3	-	1
PC13. Utilize fume extraction systems efficiently to remove emissions	2	4	-	1
PC14. Create efficient work practices to eliminate waste	2	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9819
NOS Name	Work organization and management (Welding)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N3120: Preparation and assembly techniques

Description

This NOS unit is about preparing for welding activities

Scope

The scope covers the following :

- Preparing for welding activities

Elements and Performance Criteria

Preparing for welding activities

To be competent, the user/individual on the job must be able to:

- PC1.** Set up welding equipment to manufacturers' specifications including (but not limited to):
Welding polarity, Welding amperage, Welding voltage, Wire feed speed, Travel speed, Travel/electrode angles, Mode of metal transfer
- PC2.** Prepare material edges in line with specifications and drawing requirements
- PC3.** Set up and operate appropriate controls to minimize and correct distortion
- PC4.** Carry out appropriate procedures to control heat input

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Interpretation of fabrication or engineering drawings and weld symbols
- KU2.** Classification and specific uses of welding consumables including: Coding and designation of welding rods, Diameters and specific use of welding wire, Choice and preparation of welding electrodes
- KU3.** How surface contamination can influence the finished weld characteristics
- KU4.** Correct machine settings to be aligned to: Welding polarity, Welding position, Material, Material thickness, Filler material and Feed speed
- KU5.** Any fine adjustments needed to machine hardware, TIG electrode shape, wire type and diameter etc.
- KU6.** Methods of edge preparation to align with joint profile, strength, and material
- KU7.** Methods of distortion control in steels, alloys, and aluminium

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparing for welding activities</i>	15	25	-	10
PC1. Set up welding equipment to manufacturers' specifications including (but not limited to): Welding polarity, Welding amperage, Welding voltage, Wire feed speed, Travel speed, Travel/electrode angles, Mode of metal transfer	4	7	-	3
PC2. Prepare material edges in line with specifications and drawing requirements	4	6	-	3
PC3. Set up and operate appropriate controls to minimize and correct distortion	4	6	-	2
PC4. Carry out appropriate procedures to control heat input	3	6	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3120
NOS Name	Preparation and assembly techniques
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N3121: Welding materials

Description

This NOS is about preparing raw material for welding work.

Scope

The scope covers the following :

- Preparing raw material

Elements and Performance Criteria

Preparing raw material

To be competent, the user/individual on the job must be able to:

- PC1.** Use materials with consideration to their mechanical and physical properties
- PC2.** Store welding consumables correctly with reference to type, use and safety considerations
- PC3.** Select and prepare materials with reference to drawing material list
- PC4.** Select methods used in shielding the weld area from contamination
- PC5.** Select gases used for shielding and purging

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Mechanical and physical properties of: Carbon steels, Aluminium and its alloys, Stainless steels
- KU2.** How to correct the alignment of process with the material being used
- KU3.** Selection of welding consumables
- KU4.** Correct storage and handling of welding consumables
- KU5.** Terminology, characteristics, and safe use of welding and purging gases
- KU6.** Effects of welding on the structure of the material

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparing raw material</i>	15	25	-	10
PC1. Use materials with consideration to their mechanical and physical properties	3	5	-	2
PC2. Store welding consumables correctly with reference to type, use and safety considerations	3	5	-	2
PC3. Select and prepare materials with reference to drawing material list	3	5	-	2
PC4. Select methods used in shielding the weld area from contamination	3	5	-	2
PC5. Select gases used for shielding and purging	3	5	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3121
NOS Name	Welding materials
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

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ASC/N3122: SMAW (111) and GMAW (135) Process

Description

This NOS is about performing SMAW and GMAW Process

Scope

The scope covers the following :

- Performing SMAW and GMAW Process

Elements and Performance Criteria

Performing SMAW and GMAW Process

To be competent, the user/individual on the job must be able to:

- PC1.** Make welded joints in relation to international specifications
- PC2.** Interpret welding terminology to complete task to specification
- PC3.** Perform welding of carbon steel material in all positions (except vertical down) on pipe and plates deposit single sided full penetration root pass welds
- PC4.** Deposit full penetration butt and fillet welds on pipe and plate
- PC5.** Perform stop/starts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Drawing weld symbol interpretation
- KU2.** Weld positions, weld angles and travel speeds
- KU3.** The techniques for efficient stops/starts
- KU4.** The techniques utilised to deposit single sided root penetration welds
- KU5.** The techniques utilised to deposit defect free butt and fillet welds

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performing SMAW and GMAW Process</i>	15	25	-	10
PC1. Make welded joints in relation to international specifications	3	5	-	2
PC2. Interpret welding terminology to complete task to specification	3	5	-	2
PC3. Perform welding of carbon steel material in all positions (except vertical down) on pipe and plates deposit single sided full penetration root pass welds	3	5	-	2
PC4. Deposit full penetration butt and fillet welds on pipe and plate	3	5	-	2
PC5. Perform stop/starts	3	5	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3122
NOS Name	SMAW (111) and GMAW (135) Process
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

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ASC/N3123: FCAW-G (136) Process

Description

This NOS unit is about performing FCAW-G process.

Scope

The scope covers the following :

- Performing FCAW-G process

Elements and Performance Criteria

Performing FCAW-G process

To be competent, the user/individual on the job must be able to:

- PC1.** Make welded joints in relation to international specifications
- PC2.** Interpret welding terminology to complete task to specification
- PC3.** Perform welding on carbon steel material in all positions (except vertical down) on pipe and plate
- PC4.** Perform stop/starts
- PC5.** Deposit full penetration butt and fillet welds on pipe and plate

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Drawing weld symbol interpretation
- KU2.** Weld positions, weld angles and travel speeds
- KU3.** The techniques for efficient stop/starts
- KU4.** The techniques utilised to deposit defect free butt and fillet welds

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performing FCAW-G process</i>	15	25	-	10
PC1. Make welded joints in relation to international specifications	3	5	-	2
PC2. Interpret welding terminology to complete task to specification	3	5	-	2
PC3. Perform welding on carbon steel material in all positions (except vertical down) on pipe and plate	3	5	-	2
PC4. Perform stop/starts	3	5	-	2
PC5. Deposit full penetration butt and fillet welds on pipe and plate	3	5	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3123
NOS Name	FCAW-G (136) Process
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N3124: GTAW (141) Process

Description

This NOS unit is about performing GTAW process.

Scope

The scope covers the following :

- Performing GTAW process

Elements and Performance Criteria

Performing GTAW process

To be competent, the user/individual on the job must be able to:

- PC1.** Make welded joints in relation to international specifications
- PC2.** Interpret welding terminology to complete task to specification
- PC3.** Perform welding on carbon steel, aluminium sheet, and stainless steel sheet material in all positions (except vertical down) on pipe and plate
- PC4.** Perform stop/starts
- PC5.** Deposit full penetration butt and fillet welds on pipe and plate
- PC6.** Deposit utilising a single pass on stainless steel and aluminium sheet, root and capping pass combination

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Drawing weld symbol interpretation
- KU2.** Weld positions, weld angles and travel speeds
- KU3.** The techniques for efficient stops/starts
- KU4.** The techniques utilised to deposit defect free butt and fillet welds

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performing GTAW process</i>	15	25	-	10
PC1. Make welded joints in relation to international specifications	3	5	-	2
PC2. Interpret welding terminology to complete task to specification	2	3	-	1
PC3. Perform welding on carbon steel, aluminium sheet, and stainless steel sheet material in all positions (except vertical down) on pipe and plate	3	5	-	2
PC4. Perform stop/starts	1	2	-	1
PC5. Deposit full penetration butt and fillet welds on pipe and plate	3	5	-	2
PC6. Deposit utilising a single pass on stainless steel and aluminium sheet, root and capping pass combination	3	5	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3124
NOS Name	GTAW (141) Process
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

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ASC/N3125: Finishing, quality assurance, and testing

Description

This NOS unit is about performing activities related to finishing, quality assurance, and testing on the workpiece.

Scope

The scope covers the following :

- Performing finishing, quality assurance, and testing activities

Elements and Performance Criteria

Performing finishing, quality assurance, and testing activities

To be competent, the user/individual on the job must be able to:

- PC1.** Produce welds to meet drawing and legislative specifications
- PC2.** Recognize weld defects and take appropriate action to rectify them
- PC3.** Utilize correct techniques to ensure weld metal cleanliness is maintained
- PC4.** Dress welds using wire brushes, scrapers, chisels, etc.
- PC5.** Check completed work against drawing requirements to reflect accuracy, square and flatness where necessary
- PC6.** Carry out basic non-destructive testing and be familiar with more advanced testing methods
- PC7.** Complete pressure vessels capable of withstanding hydrostatic pressure testing
- PC8.** Be creative in weld bead sequencing and weld bead tie ins to achieve uniform aesthetic appearances for faster finishing
- PC9.** Develop efficient welding procedures to reduce rework and waste during finishing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** International specifications for the control of weld quality
- KU2.** Specific terminology used in the welding industry
- KU3.** Imperfections/defects that may occur during welding
- KU4.** Importance of weld metal cleanliness in weld quality
- KU5.** A range of destructive and non-destructive testing
- KU6.** Welder certification test coupons in accordance with international standards

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Performing finishing, quality assurance, and testing activities</i>	30	50	-	20
PC1. Produce welds to meet drawing and legislative specifications	3	5	-	2
PC2. Recognize weld defects and take appropriate action to rectify them	4	6	-	3
PC3. Utilize correct techniques to ensure weld metal cleanliness is maintained	3	5	-	3
PC4. Dress welds using wire brushes, scrapers, chisels, etc.	4	6	-	2
PC5. Check completed work against drawing requirements to reflect accuracy, square and flatness where necessary	3	6	-	2
PC6. Carry out basic non-destructive testing and be familiar with more advanced testing methods	4	6	-	2
PC7. Complete pressure vessels capable of withstanding hydrostatic pressure testing	3	6	-	2
PC8. Be creative in weld bead sequencing and weld bead tie ins to achieve uniform aesthetic appearances for faster finishing	3	5	-	2
PC9. Develop efficient welding procedures to reduce rework and waste during finishing	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N3125
NOS Name	Finishing, quality assurance, and testing
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Metal Joining
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

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6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9819.Work organization and management (Welding)	30	50	0	20	100	10
ASC/N3120.Preparation and assembly techniques	15	25	0	10	50	10
ASC/N3121.Welding materials	15	25	0	10	50	10
ASC/N3122.SMAW (111) and GMAW (135) Process	15	25	0	10	50	25
ASC/N3123.FCAW-G (136) Process	15	25	0	10	50	10
ASC/N3124.GTAW (141) Process	15	25	0	10	50	15
ASC/N3125.Finishing, quality assurance, and testing	30	50	0	20	100	20
Total	135	225	0	90	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.